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Vector Model

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

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What is Data?

Collection of **data objects** and their **attributes**

An **attribute** is a **property** or characteristic of an object

- ▶ Examples: eye color of a person, temperature of a room, etc.
- ▶ Attribute is also known as **variable**, **field**, **characteristic**, or **feature**

A collection of attributes describe an object

- ▶ Object is also known as record, point, case, sample, entity, or instance

Attributes and Objects

Attributes

Objects

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attribute Values

Attribute **values** are **numbers or symbols** assigned to an attribute

The same attribute can be mapped to different attribute values

- ▶ Depends on the measurement scale
- ▶ Example: **height can be** measured **in feet or meters**

Different attributes can be mapped to the same set of values

- ▶ Example: Attribute values for ID and age are integers
- ▶ But properties of attribute values can be different
- ▶ ID has no limit, but age has a maximum and minimum value

Features, Patterns and Classifiers

Feature is any distinctive aspect, quality or characteristic

- Features may be symbolic (i.e., color) or numeric (i.e., height)

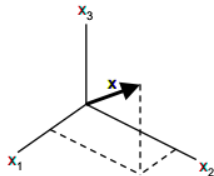
The combination of d features is represented as a d -dimensional column vector called a feature vector

- The d -dimensional space defined by the feature vector is called feature space
- Objects are represented as points in feature space. This representation is called a scatter plot

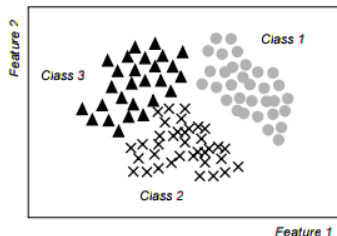
Features, Patterns and Classifiers

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

Feature vector



Feature space (3D)



Scatter plot (2D)

Pattern is a composite of traits or features characteristic of an individual

- ▶ In classification, a pattern is a pair of variables $\{x, \omega\}$ where
- ▶ x is a collection of observations or features (feature vector)
- ▶ ω is the concept behind the observation (label)

Features, Patterns and Classifiers

What makes a “good” feature vector?

- ▶ The quality of a feature vector is related to its ability to discriminate examples from different classes
- ▶ Examples from the same class should have similar feature values
- ▶ Examples from different classes have different feature values



“Good” features



“Bad” features

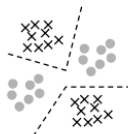
Features, Patterns and Classifiers



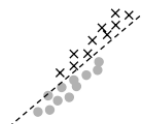
Linear separability



Non-linear separability



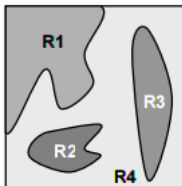
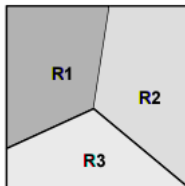
Multi-modal



Highly correlated features

Classifiers

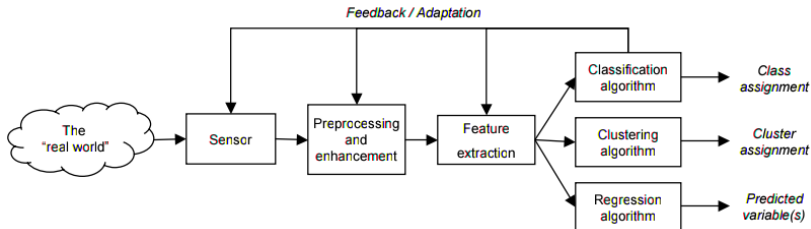
- ▶ The **goal of a classifier** is to **partition feature space** into class-labeled decision regions
- ▶ **Borders between decision regions** are called **decision boundaries**



Components of a Pattern Recognition System

A typical pattern recognition system contains

- ▶ A sensor
- ▶ A preprocessing mechanism
- ▶ A feature extraction mechanism (manual or automated)
- ▶ A classification or description algorithm
- ▶ A set of examples (training set) already classified or described



Can I eat this mushroom?



I don't know what type it is – I've never seen it before.
Is it edible or poisonous?

Can I eat this mushroom?

Suppose we're given examples of edible and poisonous mushrooms

edible



poisonous



Can we learn a model that can be used to classify other mushrooms?

Representing Instances using Feature Vectors

We need some way to represent each instance

One common way to do this: use a fixed-length vector to represent features (a.k.a. attributes) of each instance

Also represent class label of each instance

<i>cap-shape</i>	<i>cap-surface</i>	<i>cap-color</i>	<i>bruises</i>	<i>odor</i>	<i>class</i>
$\mathbf{x}^{(1)} = \langle \text{bell, fibrous, gray, false, foul, ...} \rangle$					$y^{(1)} = \text{edible}$
$\mathbf{x}^{(2)} = \langle \text{convex, scaly, purple, false, musty, ...} \rangle$					$y^{(2)} = \text{poisonous}$
$\mathbf{x}^{(3)} = \langle \text{bell, smooth, red, true, musty, ...} \rangle$					$y^{(3)} = \text{edible}$
$\vdots$					$\vdots$

Standard Feature Types

Nominal (including Boolean)

- ▶ no ordering among possible values
- ▶ e.g. color red, blue, green (vs. color = 1000 Hertz)

Linear (or ordinal)

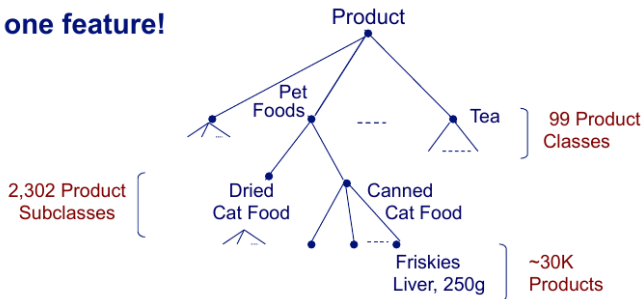
- ▶ possible values of the feature are totally ordered
- ▶ e.g. size small, medium, large ← discrete
- ▶ weight [0...500] ← continuous

Hierarchical

- ▶ possible values are partially
- ▶ ordered in an ISA hierarchy

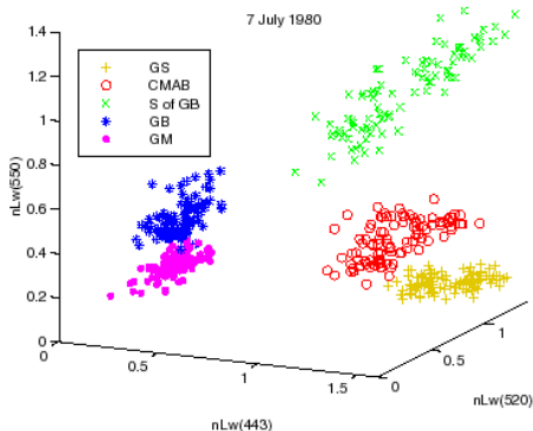
Feature Hierarchy

Structure of one feature!



Feature Space

We can think of each instance as representing a point in a d -dimensional feature space where d is the number of features



Feature Representation

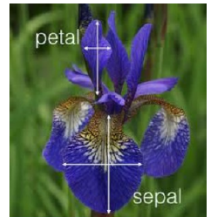
Another view of the feature-vector representation: a single database table

	feature 1	feature 2	...	feature d	class
instance 1	0.0	small		red	true
instance 2	9.3	medium		red	false
instance 3	8.2	small		blue	false
...					
instance n	5.7	medium		green	true

Feature Vector Representation

Feature Vector representations arise naturally when characterizing homogeneous entities with well-defined measurable quantities

- ▶ datum consists of measurements of well identified properties
- ▶ there is no doubt about which measurement is which
- ▶ measurement can be placed in a vector of
 - ▶ fixed dimension
 - ▶ known coordinates



Iris Dataset

- 1) sepal length
- 2) sepal width
- 3) petal length
- 4) petal width

Limits of Feature Vector Representation

Feature Vector models are hard to use when:

1. It is not always obvious how to determine satisfactory features or they are inefficient for learning purposes
 - ▶ experts cannot define features in a straightforward way
 - ▶ features consist of both numerical and categorical variables
 - ▶ missing or inhomogeneous data

Feature Extraction Inhomogeneous: the entries differ significantly in structure, type, or scale. Data points may include numerical values, categorical labels, text, images, or other formats.

2. Symmetries and invariances make it so that the identity of the measurement must be inferred from the context (i.e. image analysis)

Contextual Pattern Recognition



Limits of Feature Vector Representation

4. Measurables are overabundant, resulting in very high-dimensional data
(curse of dimensionality)

Dimensionality Reduction

5. Entities are not homogeneous and vary in complexity, are described in terms of structural properties, such as parts and relations between parts

Structural Pattern Recognition